

VERSION 2.0 — B.TECH FINAL-YEAR PROJECT

The system that knows *how you learn.*

A cognitive-load scheduler, reframed as a closed loop: what to learn, how to learn it, and when — each pillar built and judged on its own terms.

THE STRUCTURAL FAILURE

Three systems that don't talk to each other.

01. Fixed Formats

One generator, one format, per topic – regardless of whether the learner is struggling or already fluent.

02. Blind Discovery

Trend surfacing runs one-way. It has no idea which topics the scheduler already knows you're behind on.

03. The Disconnect

What, How, and When exist as three separate agents today – not one loop that learns from itself.

THE ARCHITECTURE

What to learn. How to learn it. When to learn it.

01 – WHAT

Trend Engine

Hacker News, Reddit, and NewsAPI signal, mapped into candidate learning topics.

02 – HOW

Material Generators

Roadmap, course, flashcard, assessment, cheatsheet, and story agents turn a topic into content.

03 – WHEN

Scheduling Engine

Content placed against cognitive load, energy windows, and deadlines – deterministic, rule-based.

Only **When** is closed-form enough to research without human subjects. **What** and **How** are the engineering track – proof of orchestration skill, not proof by experiment.

COMPLETE TRANSPARENCY

No hidden algorithm. No black box.

Axiom doesn't quietly "optimize." When two tasks want the same slot, it applies strict rules and shows exactly how it chose — this is already running in production today.

HOW AXIOM HANDLED A CONFLICT AT 2:00 PM

1. Cognitive cost of "Write Report": $E_{task} = 8.4$ (high)
2. Bandwidth available at 1:00 PM: low (post-lunch trough)
3. **RULE: high- E_{task} tasks do not enter low-bandwidth slots.**
4. Next high-bandwidth slot: 2:00 PM
5. **CONFLICT: "Team Sync" already occupies 2:00 PM**
6. **RULE: on clash, higher-fitness task keeps the slot.**
7. → **ACTION: place "Write Report" at 2:00 PM.**
8. → **ACTION: move "Team Sync" to 3:30 PM.**

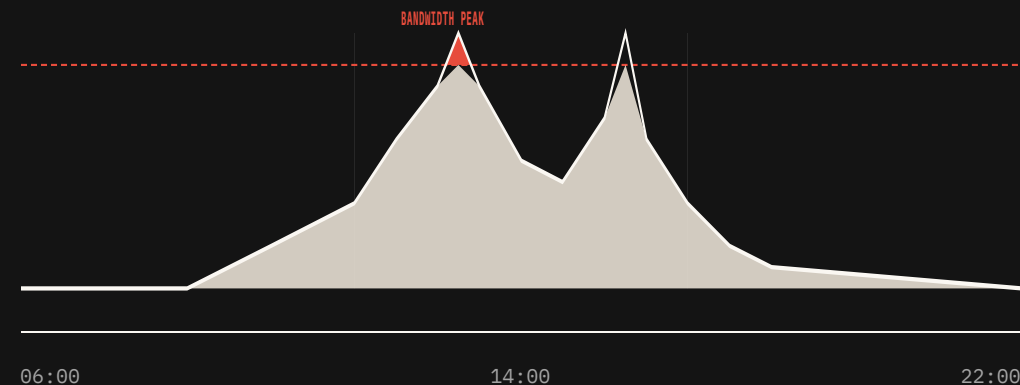
RESEARCH CORE — THE WHEN PILLAR

Etask, fitness, and a greedy loop we can now question.

```
Etask          = difficulty × durationWeight × deadlineUrgency ×  
|typeMultiplier| × priorityWeight  
Fitness(slot) = bandwidth - Etask + productiveBonus +  
switchPenalty + deadlineBonus
```

runSchedulingAlgorithm places tasks one at a time, greedily, in sorted order.

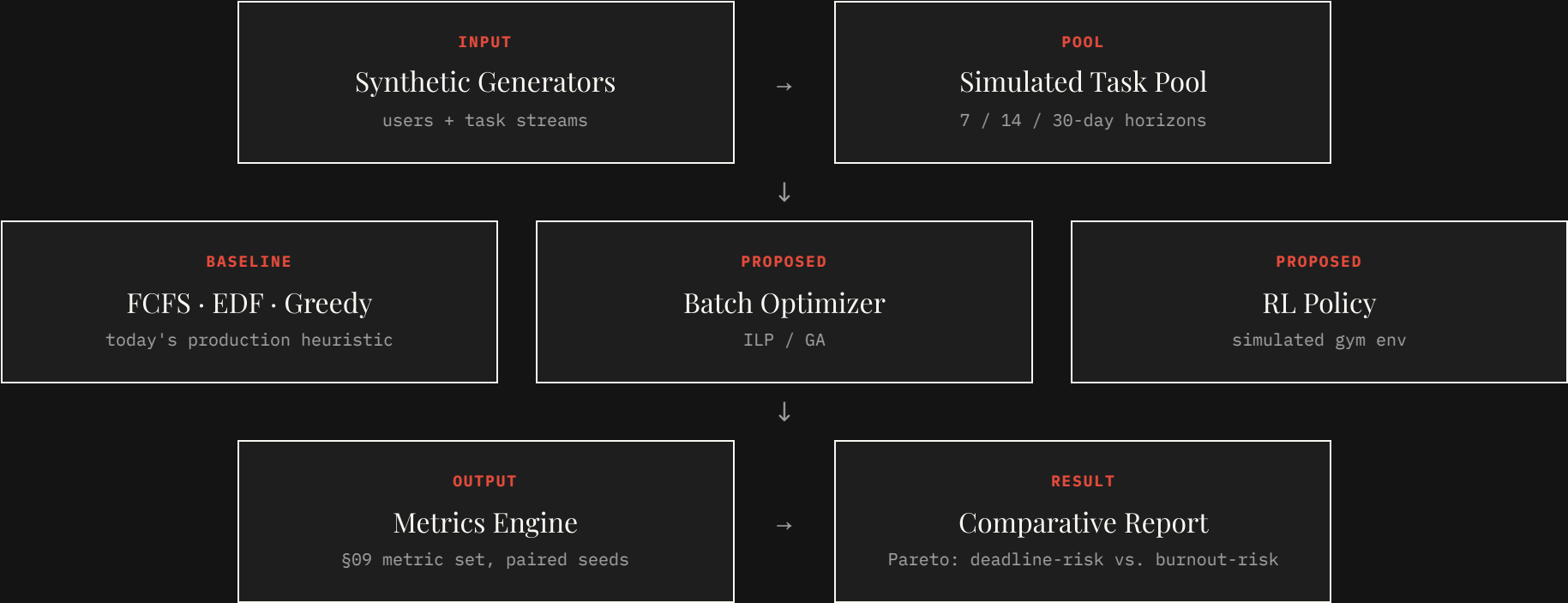
Order-dependent. Single scalar score. **Research question:** how much schedule quality is left on the table by a greedy, one-pass heuristic – and what trade-off between deadline risk and burnout risk does collapsing everything into one number hide?



The ultradian bandwidth curve behind `Fitness(slot)` – `generateDefaultBandwidthCurve()`. Two peaks, two troughs, per day.

THE RESEARCH HARNESS

Same formulas, entirely simulated data.



NO HUMAN SUBJECTS. NO IRB. NO SURVEYS.

Every research claim is computed against synthetic data.

01

SYNTHETIC USERS

Wake/sleep, peak-focus windows, energy profile – sampled from parameterised distributions.

02

SYNTHETIC TASKS

Poisson arrivals; randomised difficulty, duration, deadline, type, priority.

03

ALGORITHM BANK

FCFS, EDF, current greedy heuristic, proposed ILP / GA / RL candidates.

04

METRICS + PARETO

Feasibility, deadline-miss, CL variance, burnout rate, solve time – paired seeds, significance tested.

NOT RESEARCH — DEMONSTRATED ENGINEERING

Where What and How prove the trending stack.

WHAT

Multi-Agent Trend Pipeline

Fetch-agent → dedupe/cluster-agent → LLM relevance-filter → summarizer → mastery-gap re-ranker.

WHAT

Tool-Calling Source Selection

An LLM decides which source to query per topic domain, instead of always querying all three.

HOW

Self-Refinement Loop

Critic LLM scores each draft against a rubric; regenerate below threshold. An automated quality gate, not a human one.

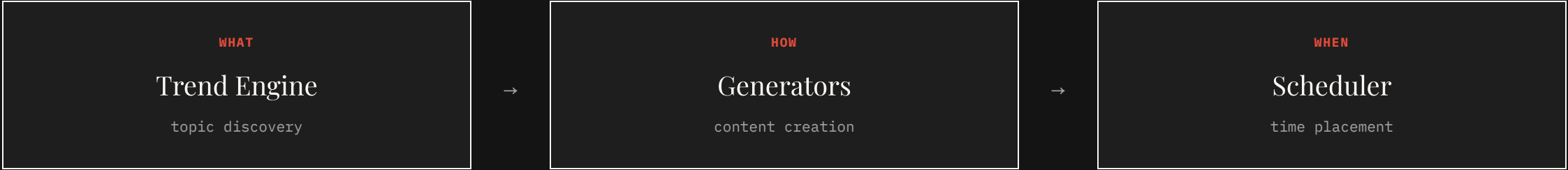
HOW

Model Routing + Adaptive ICL

Cheap model for flashcards, strong model for course slides; few-shot examples pulled from the user's own history.

END TO END

Calibration flows back into What and How, not just When.



🔗 **Mastery-weighted ranking** – completion-rate calibration feeds back into topic ranking.

🔗 **Performance-aware generation** – the same signal adapts content difficulty and format.

SIXTEEN WEEKS

One loop, phased.

WK 1-2	Formalise	Literature grounding; write the scheduler as a formal constrained optimization.
WK 3-5	Build the harness	Synthetic generators, baseline algorithms, metrics engine.
WK 6-9	Implement	ILP/GA batch optimizer; RL policy if time allows.
WK 10-12	Experiment	Run all algorithms across seeds/horizons; statistical tests, Pareto plots.
WK 13-14	Integrate + write	Feature-flag the best optimizer into the product; draft the paper.
WK 15-16	Showcase + defend	What/How engineering features; final report and viva.

READY TO BUILD

One loop. Three pillars. Sixteen weeks.

Approve – When as the sole research/evaluation track, no human subjects required.

Approve – What / How as an engineering showcase: orchestration, adaptive LLM technique.

Commit – 16-week phased plan, scheduler-first, showcase features as stretch scope.

AXIOM – COGNITIVE LEARNING OS